Mutaher Affan

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Education

Bachelor of Science (BS), Computer Science Jan 2018 – Jan 2022

University of Sindh - 3.38 CGPA

- Developed desktop applications using Java Swing and built web/RESTful backends using Spring Boot and Django REST Framework.
- Gained hands-on experience in programming with Python, Swagger/OpenAPI, and other related technologies.

Technical Skills

Languages: Java, SQL
Frameworks: Spring Boot, Spring Cloud, Spring Security, Spring Data JPA, Hibernate, JUnit 5
Technologies: Docker, RabbitMQ, Redis, API Gateways, Swagger/OpenAPI, Microservices, Kubernetes, JIRA, Git, GitLab
Databases: MS SQL Server, IBM DB2
Tools: Selenium, k6 Load Testing, Swagger Editor, ELK Stack
Methodologies: REST API Design, Object-Oriented Programming (OOP), Software Documentation, Unit Testing, Agile

Experience

Systems Limited Feb 2026 - Present

Consultant - App Development *Karachi, Pakistan*

- Working as a consultant on enterprise Identity and Access Management (IAM) solutions, supporting authentication and authorization services built on Keycloak Managed Services.
- Responsible for managing, configuring, and maintaining Keycloak environments, ensuring secure, scalable, and highly available identity services for enterprise applications.
- Implemented and supported authentication and authorization workflows using OAuth 2.0, OpenID Connect (OIDC), and role-based access control (RBAC) to secure application access.
- Collaborated with cross-functional teams to troubleshoot identity-related issues, optimize security configurations, and streamline user access management.
- Tech Stack: Keycloak, OAuth 2.0, OpenID Connect (OIDC), RBAC, Java, REST APIs, Docker, Git.

Paysys Labs Dec 2022 - Feb 2026

Senior Software Engineer *Karachi, Pakistan*

- Evolved from Associate Software Engineer to Senior Software Engineer, playing a key role in the design, development, and optimization of complex software systems and middleware services.
- Led and contributed to the enhancement of scalable backend systems, ensuring high performance, security, and seamless integrations across multiple projects.
- Collaborated with cross-functional teams to define and implement service architecture, performance tuning, and best practices for API integrations.
- Tech Stack: Java, Spring Boot, Hibernate, Docker, Redis, Junit5 with Mockito, SQL Server, RabbitMQ, Nginx.

Teresol Pvt. Ltd. Mar 2022 - Dec 2022

Software Design Engineer *Karachi, Pakistan*

- Refactored legacy systems using Java Quarkus Framework, improving authentication and authorization mechanisms for core banking applications.
- Integrated new technologies and ensured compliance with security standards for financial systems.
- Tech Stack: Java, Quarkus, DB2, Docker, Git, ELK Stack.

GBS Feb 2021 - May 2021

Django Developer *Hyderabad, Pakistan*

- Designed and developed RESTful APIs using Django, ensuring seamless integration with ReactJS for efficient backend/frontend communication.
- Focused on scalability, performance, and optimal user experience while maintaining a secure and maintainable codebase.
- Tech Stack: Django, ReactJS, SQL Server.

Projects

CelcomDigi – Identity and Access Management (IAM) *Feb 2026 – Present*

Keycloak Managed Services | *Keycloak, OAuth 2.0, OpenID Connect (OIDC), RBAC, Java, REST APIs, Docker*

- Managed enterprise Identity and Access Management (IAM) services for CelcomDigi using Keycloak Managed Services, ensuring secure authentication and authorization across multiple applications.
- Configured and maintained Keycloak realms, clients, roles, groups, and user access policies while supporting OAuth 2.0 and OpenID Connect (OIDC) authentication flows.
- Troubleshot authentication, authorization, and token-related issues, collaborating with application teams to ensure secure and seamless user access.
- Supported identity lifecycle management, access control, and security best practices to maintain high availability and compliance of IAM services.

IPS Integration (Eth-Switch) – ISO 20022 Real-Time Payments *Jul 2025 – Present*

ISO 20022 Integration for P2P/P2M Flows | *Java, Spring Boot, REST APIs, ISO 20022, JAXB*

- Integrated the IPS platform from BPC using the ISO 20022 standard, mapping and validating PACS messages for P2P/P2M flows between Open-Connect and bank cores.
- Designed and exposed REST request APIs with JSON payloads to simplify bank integrations and accelerate onboarding, while preserving ISO 20022 semantics.
- Implemented schema validation (XSD), correlation/trace IDs for observability, and performance tuning to ensure secure and efficient bank integration.
- Collaborated with Eth-Switch and BPC teams in production-like environments to ensure reliable, secure, and well-documented integrations.

PayVay Digital Services (PSO – CERISMA) – Architecture & Platform Services *Nov 2024 – Present*

Microservice Architecture Design and ISO Integration | *Java, Spring Boot, Spring Cloud, Hibernate, MS SQL Server, Docker, Prometheus, Grafana*

- Designed and led a PSO-grade microservice architecture, defining service boundaries, shared contracts, and resiliency patterns.
- Implemented an API Gateway with Spring Cloud Gateway and layered security using Spring Security and mTLS, standardizing request/response models, error handling, and audit logging.
- Developed ISO 8583 interfaces, including robust field mapping, MAC validation, and reversal/chargeback flows, integrating them with ISO 20022 services and internal canonical models.
- Established best practices such as versioned APIs, OpenAPI documentation, webhook/eventing patterns, and CI code-quality gates (SonarLint).

NBP Digital Services – REST API Development *Dec 2023 – Present*

Digital Banking App Enhancement | *Spring Boot, Hibernate, Docker, MS SQL Server, Spring Cloud*

- Collaborated with cross-functional teams to develop and maintain National Bank of Pakistan's digital banking application.
- Contributed to middleware development, optimizing data flow, and enhancing application performance for high-volume banking transactions.
- Implemented secure REST APIs and backend enhancements while ensuring seamless integration across enterprise services.

CDNS Digital Services – REST API and Middleware Development *Jan 2023 – Present*

REST API and Middleware Development | *Spring Boot, Hibernate, Docker, MS SQL Server, Git*

- Developed RESTful APIs and middleware services to enable secure communication between CDNS digital platforms.
- Enhanced API scalability, performance, and security while streamlining enterprise data exchange.
- Collaborated within agile teams to deliver production-ready solutions aligned with business requirements.

User Authentication and Authorization (AAA) *Mar 2022 – Dec 2022*

Core Banking Authentication Modernization | *Java, Quarkus, DB2, Docker, ELK Stack, Kubernetes*

- Modernized the authentication and authorization platform for Bank Al Habib by refactoring legacy applications using Java Quarkus.
- Enhanced security, maintainability, and deployment efficiency through containerization and cloud-native practices.
- Integrated ELK Stack for centralized logging and monitoring while leveraging Harbor for container image management.

Certifications

Java Basic Oct 2024

HackerRank — Online

- Demonstrated proficiency in core Java concepts such as classes, data structures, inheritance, and exception handling.
- Gained hands-on experience with Java 8, Java 11 and Java 17 syntax and features.
- Showcased strong problem-solving and programming fundamentals in Java.

Spring Boot 3, Spring 6, and Hibernate for Beginners Aug 2019

Udemy — Online

- Gained hands-on experience building modern Java applications with Spring Boot and Hibernate.
- Developed RESTful APIs and implemented dependency injection with Spring Framework.
- Learned persistence and ORM concepts using Hibernate and JPA integration.

JUnit 5 and Mockito Jul 2023

Udemy — Online

- Acquired expertise in unit testing and test-driven development (TDD) using JUnit 5 and Mockito.
- Developed skills in dependency mocking and writing effective test cases.
- Enhanced code quality through automation and robust test coverage practices.

Dockerization Certificate Oct 2023

Paysys Labs — Online

- Learned to containerize applications using Docker for development and deployment.
- Built and managed Docker images and containers for various environments.
- Gained practical experience orchestrating multi-container setups with Docker Compose.

Python (Basic) Jan 2023

HackerRank — Online

- Completed the Python Basic Certification covering core programming fundamentals.
- Practiced Python concepts including data types, loops, and conditional statements.
- Demonstrated problem-solving proficiency using Python programming.

Programming Essentials in Python Mar 2020

OpenEDG Python Institute — Online

- Learned essential Python programming concepts such as variables, loops, and functions.
- Developed understanding of object-oriented programming principles in Python.
- Built a strong foundation for Python-based software development.